

ANJANA KRISHNAMOORTHY

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EDUCATION

Masters in Computer Science

California State University, Dominguez Hills (CSUDH) GPA: 3.85/4.0

May 2026

Carson, CA

Courses: Design and Analysis of Algorithms, Data Structures and Algorithms, Operating Systems, Computer Architecture, Database Management systems, Programming Languages, Software Engineering, Theory of Computation, Object-Oriented Programming and Design, Advanced Software Engineering, Software Project Planning, Software Quality Assurance, Human-Computer Interaction.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C++, C, Swift, R, SQL, SAS

Data Science & Analytics: Pandas, NumPy, Seaborn, Matplotlib, Scikit-learn, TensorFlow, PyTorch, HuggingFace, Excel, Google Sheets

Statistical Modeling & Machine Learning: Logistic Regression, CNN, XGBoost, Predictive Modeling, Contrastive Learning, NLP, Explainable AI

Analytics Tools & Platforms: R Shiny, SAS Base, SDTM, ADaM, TLFs, Jupyter Notebook, Google Colab, Flask

Databases: MySQL, MongoDB, SQLite, PostgreSQL

PROJECTS

Modular Automated Essay Scoring System :

January 2025 - April 2025

Python, PyTorch, BERT, HuggingFace, Contrastive Learning, ASAP Dataset

- Designed a modular deep learning pipeline for essay scoring using three specialized models: BERT-based holistic scorer, contrastive coherence evaluator, and grammar module.
- Engineered a coherence evaluator trained with synthetically shuffled texts, enabling detection of discourse-level organization via contrastive learning.
- Integrated model outputs through weighted averaging and a 2-layer meta-model (MLP) to produce final scores.
- Trained on the ASAP-AES dataset (12,976 essays), achieving QWK: 0.96 and RMSE: 0.002 on test set evaluations.
- Enabled interpretable score breakdowns for grammar, structure, and coherence, supporting educational feedback use cases.

Medtoureasy – Healthcare Risk Analysis :

March 2024 - April 2024

Python, Pandas, Seaborn, Matplotlib, Excel

- Constructed dashboards to track trends across clinical, demographic, and treatment data, supporting data-driven planning.
- Refined preprocessing workflows for 50K+ medical records, improving data quality and model input readiness.
- Engineered a logistic regression model for mortality risk prediction with 98% accuracy on internal validation.
- Collaborated with healthcare domain experts to validate outputs and derive actionable insights.

Pfizer MDR Dashboard – Clinical Data Visualization :

September 2024 - December 2024

C++, Autodesk Inventor COM API, BNF, HIPAIR input grammar

- Developed a standalone C++ executable that extracted 2D boundary geometries from Inventor's .ipt and .iam files and converted them to HIPAIR's BNF-compliant format.
- Retrieved 12 line segments and 4 arc definitions from two custom 3D part files, preserving spatial structure for accurate simulation.
- Analyzed part orientation and extrusion depth to ensure correct inter-part assembly behavior in the HIPAIR environment.
- Enabled support for arcs, bidirectional sketching, and hard-coded assembly constraints for 2 test mechanisms in simulation.

EXPERIENCE

Computer Science Scholar

February 2025 - Present

Toro Auxiliary Partners - CSUDH

Compton, CA

Swift, Swift Playgrounds (iPadOS), Sphero Edu, Lesson Design, Communication

- Delivered Swift programming workshops to 100+ K-12 students, transforming complex logic into engaging puzzles.
- Facilitated hands-on robotics sessions with Sphero bots, increasing student engagement and STEM interest by 40%.
- Designed and modified 10+ curriculum-aligned lessons for diverse age groups, promoting inclusive and adaptive learning.
- Enhanced classroom effectiveness by incorporating real-time feedback and student-centered instruction strategies.

Data Analytics Trainee

March 2024 - April 2024

Medtoureasy

India

Python, Pandas, NumPy, Seaborn, Matplotlib, Data Cleaning, Data Visualization, Statistical Analysis

- Constructed interactive dashboards to visualize real-time healthcare trends, increasing data-driven decision-making by 65%.
- Refined preprocessing pipelines to clean and transform 50K+ clinical records, boosting analysis accuracy by 45%.
- Engineered a logistic regression model for mortality prediction with 98% accuracy, supporting early risk detection.
- Collaborated with domain experts to evaluate insights and deliver clinically actionable analytics.

Python Programming Intern

February 2024 - March 2024

Oasis Infobyte

India

Python, Automation, CLI Tools, Git, VS Code

- Automated repetitive workflows using Python scripts, reducing manual effort by 25% and boosting operational efficiency by 55%.
- Troubleshoot and optimized core logic, decreasing script runtime by 30% across test cases.
- Developed 3+ custom CLI tools for data formatting and batch processing, accelerating execution time by 40%.
- Documented and organized code using Git, increasing reusability and enabling effective team collaboration.

Associate Statistical Data Scientist Intern

April 2023 - July 2023

Pfizer

India

SAS, Clinical Data Analysis, SDTM/ADaM, TLFs, Predictive Modeling, Statistical Programming

- Analyzed large-scale clinical trial datasets using SAS, delivering outputs with 99%+ accuracy aligned to FDA compliance standards.
- Designed predictive models to evaluate drug efficacy, supporting high-stakes R&D decision-making.
- Streamlined statistical programming workflows, reducing report generation time by 35% across multiple studies.
- Collaborated with senior statisticians to convert SDTM/ADaM datasets and generate TLFs for regulatory submissions.

Awards and Certificates

Research Paper (RANLP 2025): Co-authored "Multi-Agent Reinforcement Learning for Interactive Code Debugging with Human Feedback and Memory" exploring RLHF, BERT, PPO agents, and FAISS-based personalization for developer-aligned debugging.